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B.Tech. Degree III Semester Examination November 2014

ME 1303 MECHANICS OF SOLIDS (2012 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer ALL questions)

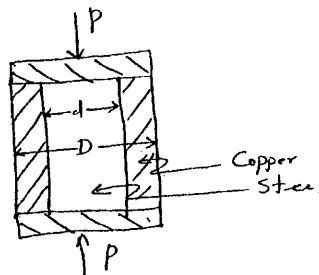
(8 × 5 = 40)

- I.
 - (a) Draw the stress strain diagram of a typical ductile material and mark the salient points.
 - (b) Differentiate between plane stress and plane strain.
 - (c) List the assumptions in torsion equation.
 - (d) What is point of contra flexure? Illustrate with a simple example.
 - (e) State and explain any one theory of failure applicable for cast iron.
 - (f) Find the ratio of width to depth (b/d) of the strongest beam obtainable from a cylinder of dia 'D'.
 - (g) What is meant by equivalent length of a column? What are its values under the typical loading conditions?
 - (h) A beam of rectangular section 100 mm wide and 240 mm deep is simply supported over a span of 4 m. What uniformly distributed load will make a central deflection of 6 mm? Take $E = 0.11 \times 10^5 \text{ N/mm}^2$

PART B

(4 × 15 = 60)

- II. A steel cylinder is inserted in a copper tube as shown. The cylinder and tube are compressed between rigid parallel plates. Find the stresses in the steel and copper and also the compressive strain. Take $P = 450 \text{ kN}$, $d = 100 \text{ mm}$, $D = 200 \text{ mm}$ For steel $E = 210 \text{ kN/mm}^2$. For copper, $E = 110 \text{ kN/mm}^2$.



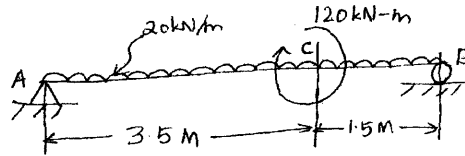
OR

- III. Calculate the thickness of metal required for a C.I pipe of 800 mm diameter, to transport water under a static head of 100 m, if the permissible tensile stress is 20 MPa.
- IV. Find the speed at which 200 kW should be transmitted through a shaft of 60 mm diameter, if the maximum permissible shear stress is 50 MPa. If the length of the shaft is 6m, find the angle of twist. Take $G = 80 \text{ GPa}$.

OR

(P.T.O.)

- V. Construct the shear force and bending moment diagram for the beam shown:



- VI. A cast iron water pipe 450 mm inner diameter and 25 mm thickness is supported at two points 10 m apart. Find the maximum stress in the metal when it is running full. Unit weight of C.I. is 72 kN/m^3 .

OR

- VII. At a certain cross section of a shaft, 100 mm in diameter, there is a bending moment of 5 kN-m and a twisting moment of 7.5 kN-m. Calculate the maximum direct stress induced in the section and specify the position of the plane on which it acts.

- VIII. A beam of size $300 \times 150 \text{ mm}^2$ is freely supported over a span of 5m. It carries two point loads of 30 kN and 40 kN at 1m and 3.75 m from the left support. Find the position and location of maximum deflection.

OR

- IX. A vertical steel strut of 40 mm diameter and 3 m long, has pinned ends. Taking $E = 215 \text{ GPa}$, calculate the crippling axial load using Euler's formula. Also find the maximum central deflection on buckling if the yield stress is 310 N/mm^2 is reached. Neglect the effect of axial thrust.
